GATE ME 90 Days Challenge

Day 62/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Manufacturing	Metrology & CIM	39 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on limits, fits and tolerances (IT grades, hole basis, fundamental deviations), limit gauges and tolerance stacks, sine bar and slip gauges, comparators, interferometry and optical flats, surface roughness, geometric tolerances (MMC), and CNC part programming, BLU and CAD transformations, CAPP. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

The hole-basis system of fits is preferred in most industries because

(A) shafts are easier to measure than holes

(B) it always gives interference fits

(C) holes are produced with fixed-size tools (drills, reamers, broaches) and checked with plug gauges, so keeping the hole size constant needs fewer tools and gauges

(D) the fundamental deviation of every shaft is zero

Q2 NAT

A fit is specified as 50 H7/f6. For this size IT7 = 25 µm and IT6 = 16 µm, and the fundamental (upper) deviation of the f shaft is −25 µm. The maximum clearance of the fit (in µm) is _____

Q3 MCQ

Which one of the following combinations gives a transition fit?

(A) H7/k6

(B) H7/g6

(C) H7/s6

(D) H8/d9

Q4 NAT

A 100 mm sine bar is to be set to an angle of 35°. The height of the slip-gauge pile to be placed under one roller (in mm, round off to two decimal places) is _____

Q5 MCQ

A linear variable differential transformer (LVDT) works on the principle of

(A) change of resistance of a strained wire

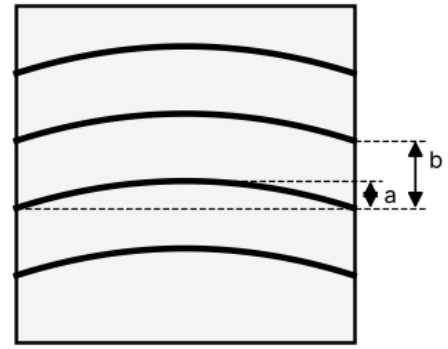
(B) change of mutual inductance between a primary coil and two secondary coils as a soft-iron core moves

(C) change of capacitance between two plates

(D) generation of an EMF by a piezoelectric crystal

Q6 NAT

A surface is tested with an optical flat under monochromatic light of wavelength 0.6 µm. The fringes are curved as shown, and the bow of each fringe is 0.4 of the fringe spacing. The out-of-flatness of the surface (in µm, round off to two decimal places) is _____



fringes seen through the optical flat (a/b = 0.4)

Q7 MCQ

Which statement about surface roughness parameters is correct?

(A) Ra is the maximum peak-to-valley height within the sampling length

(B) Ra is always larger than Rz for the same surface

(C) the sampling (cut-off) length has no effect on the measured roughness

(D) Ra is the arithmetic mean of the absolute deviations of the profile from the mean line over the sampling length

Q8 NAT

A turning tool with a nose radius of 0.6 mm is used at a feed of 0.15 mm/rev. The ideal (geometric) peak-to-valley surface roughness (in µm, round off to two decimal places) is _____

Q9 MCQ

A hole of size 20.0–20.2 mm has a position tolerance of Ø0.1 mm at maximum material condition (MMC). If the hole is actually produced at 20.15 mm, the position tolerance available is

(A) Ø0.10 mm

(B) Ø0.25 mm

(C) Ø0.30 mm

(D) Ø0.05 mm

Q10 NAT

A gauge block is measured with a He-Ne laser interferometer (λ = 632.8 nm). As the moving mirror travels from one face of the block to the other, 50 000 fringes are counted. The length of the block (in mm, round off to two decimal places) is _____

Q11 MCQ

In ISO CNC programming, which one of the following codes is correctly matched with its function?

(A) G00 – linear interpolation at the programmed feed

(B) M03 – spindle stop

(C) M06 – tool change

(D) G91 – absolute dimensioning

Q12 NAT

The table of a CNC drilling machine is driven directly (no gearing) by a stepper motor with a step angle of 1.8° through a lead screw of 4 mm pitch. The number of pulses needed to move the table by 75 mm is _____

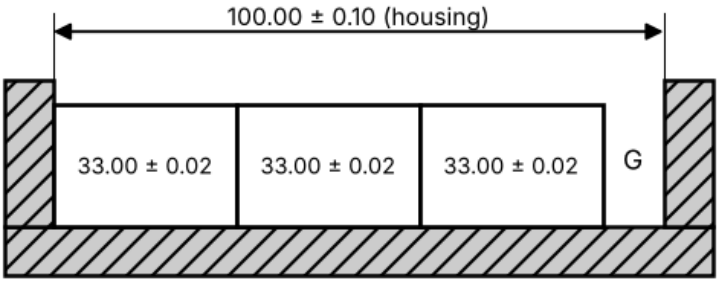
Q13 MCQ

In 2-D CAD, the point (3, 4) is scaled by a factor of 2 in both directions about the fixed point (1, 1). Its new position is

(A) (6, 8)
(B) (4, 7)
(C) (7, 5)
(D) (5, 7)

Q14 NAT

Three spacers, each 33.00 ± 0.02 mm long, are placed side by side in a housing whose internal length is 100.00 ± 0.10 mm, as shown. The maximum possible gap G (in mm, round off to two decimal places) is _____



(not to scale)

Q15 MCQ

In computer-aided process planning (CAPP), the generative approach

(A) creates a new process plan from part features using decision logic, formulae and a manufacturing knowledge base, without starting from a stored plan
(B) retrieves the standard plan of the part family (found by group-technology coding) and edits it
(C) is the same as the variant approach
(D) applies only to assembly planning

Answer Key

Q	Type	Answer
Q1	MCQ	(C) holes are produced with fixed-size tools (drills, reamers, broaches) and checked with plug gauges, so keeping the hole size constant needs fewer tools and gauges
Q2	NAT	66
Q3	MCQ	(A) H7/k6
Q4	NAT	57.36 (accept 57.33 to 57.39)
Q5	MCQ	(B) change of mutual inductance between a primary coil and two secondary coils as a soft-iron core moves
Q6	NAT	0.12
Q7	MCQ	(D) Ra is the arithmetic mean of the absolute deviations of the profile from the mean line over the sampling length
Q8	NAT	4.69 (accept 4.66 to 4.72)
Q9	MCQ	(B) Ø0.25 mm
Q10	NAT	15.82
Q11	MCQ	(C) M06 – tool change
Q12	NAT	3750
Q13	MCQ	(D) (5, 7)
Q14	NAT	1.16
Q15	MCQ	(A) creates a new process plan from part features using decision logic, formulae and a manufacturing knowledge base, without starting from a stored plan

Detailed Solutions

Q1 Answer: (C) holes are produced with fixed-size tools (drills, reamers, broaches) and checked with plug gauges, so keeping the hole size constant needs fewer tools and gauges

Concept: In the hole-basis system the lower deviation of the hole is zero (H hole) and different fits are obtained by changing the shaft.

Shafts can be turned or ground to any size with the same tool, but each hole size needs its own drill, reamer, broach and plug gauge. Keeping the hole fixed therefore cuts tooling and gauging cost: **(C)**.

Exam tip: The shaft-basis system (h shaft) is used when one shaft of standard bright bar carries several parts with different fits.

Q2 **Answer: 66**

Concept: H hole: lower deviation 0, upper deviation = IT grade. f shaft: upper deviation = fundamental deviation, lower = upper − IT grade.

Hole: 50.000 to 50.025 mm

Shaft: upper = 50 − 0.025 = 49.975 mm; lower = 49.975 − 0.016 = 49.959 mm

Maximum clearance = 50.025 − 49.959 = 0.066 mm = **66 µm** (minimum clearance = 25 µm)

Exam tip: For shafts a to h the fundamental deviation is the upper deviation; for j to zc it is the lower deviation.

Q3 **Answer: (A) H7/k6**

Concept: In the hole-basis system, shafts a–h give clearance fits, j–n transition fits and p–zc interference fits (roughly).

A k6 shaft overlaps the H7 hole zone, so an assembled pair may have a small clearance or a small interference: **(A)**.

g6 and d9 give clearance fits (sliding/running); s6 gives an interference (press) fit.

Exam tip: Transition fits such as H7/k6 or H7/n6 are used for gears and couplings that must be located accurately but can still be dismantled.

Q4 **Answer: 57.36 (accept 57.33 to 57.39)**

Concept: The sine bar forms the hypotenuse of a right triangle: sin θ = h/L, where L is the distance between the roller centres.

h = 100 × sin 35° = 100 × 0.57358

h = **57.36 mm**

Exam tip: Sine bars become inaccurate above about 45°, since an error in h causes an angle error dθ = dh/(L cos θ), which grows as θ increases.

Q5 **Answer: (B) change of mutual inductance between a primary coil and two secondary coils as a soft-iron core moves**

Concept: An LVDT has one primary and two secondary windings connected in series opposition, with a movable soft-iron core.

At the null position the two secondary voltages cancel; displacing the core raises one and lowers the other, so the output voltage is proportional to displacement and its phase shows the direction: **(B)**.

Exam tip: The LVDT is an electrical comparator; it is frictionless, has very high resolution and is linear over its working range.

Q6 **Answer: 0.12**

Concept: Neighbouring fringes differ in air-gap thickness by λ/2. A fringe curved by a fraction a/b of the fringe spacing shows a height error of (a/b) × λ/2.

λ/2 = 0.3 µm

Out-of-flatness = 0.4 × 0.3 = **0.12 µm**

Exam tip: Straight, parallel, equally spaced fringes mean a flat surface; curved fringes mean a convex or concave surface (pressing the flat at one edge tells which).

Q7 **Answer: (D) Ra is the arithmetic mean of the absolute deviations of the profile from the mean line over the sampling length**

Concept: Ra (CLA) = (1/L)∫|y| dx, measured from the mean line within the sampling length.

Rz (ten-point height) and Rt/Rmax are peak-to-valley type parameters, so they are always larger than Ra (for example Ra = h/4 for a triangular profile): **(D)**.

The cut-off length filters out waviness; a longer cut-off includes more waviness and changes the reading.

Exam tip: Ra is the most widely used parameter but cannot distinguish peaks from valleys; two very different surfaces can have the same Ra.

Q8 **Answer: 4.69 (accept 4.66 to 4.72)**

Concept: When the feed is small compared with the nose radius, the surface is a series of circular arcs and $h = f^2/(8r)$.

$h = 0.15^2/(8 \times 0.6) = 0.0225/4.8 = 0.0046875$ mm

h = **4.69 µm**

Exam tip: Halving the feed cuts the ideal roughness to one-quarter, while doubling the nose radius only halves it.

Q9 **Answer: (B) Ø0.25 mm**

Concept: With the MMC modifier, the stated geometric tolerance applies when the feature is at MMC, and any departure from MMC is added as a bonus tolerance.

MMC of a hole = smallest size = 20.0 mm. Departure = 20.15 − 20.0 = 0.15 mm

Available position tolerance = 0.1 + 0.15 = **Ø0.25 mm (B)**

Exam tip: The maximum bonus occurs at LMC (20.2 mm), giving Ø0.3 mm. A functional GO gauge checks size and position together at the virtual condition 20.0 − 0.1 = 19.9 mm.

Q10 **Answer: 15.82**

Concept: In a Michelson-type interferometer the light travels to the mirror and back, so each fringe corresponds to a mirror movement of λ/2.

$L = N\lambda/2 = 50\,000 \times 0.6328/2$ µm = 15 820 µm

L = **15.82 mm**

Exam tip: Laser interferometers are used to calibrate CNC machine axes; air temperature, pressure and humidity change λ and must be compensated.

Q11 **Answer: (C) M06 – tool change**

Concept: G codes are preparatory (motion and mode) functions; M codes are miscellaneous (machine) functions.

G00 is rapid positioning (G01 is linear interpolation at feed), M03 is spindle on clockwise (M05 is stop), G91 is incremental (G90 is absolute). M06 is tool change: **(C)**.

Exam tip: Also remember G02/G03 (CW/CCW arcs), G17 (XY plane), M08/M09 (coolant on/off) and M30 (end of program, rewind).

Q12 **Answer: 3750**

Concept: Basic length unit (BLU) = table movement per pulse = pitch/(steps per revolution × gear ratio).

Steps per revolution = 360/1.8 = 200, so BLU = 4/200 = 0.02 mm

Pulses = 75/0.02 = **3750**

Exam tip: The table speed is set by the pulse frequency: at 1000 pulses/s this table moves 20 mm/s.

Q13 **Answer: (D) (5, 7)**

Concept: Scaling about a point other than the origin = translate that point to the origin, scale, translate back.

Relative position: (3 − 1, 4 − 1) = (2, 3); scaled: (4, 6); back: (4 + 1, 6 + 1) = **(5, 7)**

(6, 8) is the result of scaling about the origin.

Exam tip: In homogeneous coordinates the three steps become one 3 × 3 matrix product T(1,1)·S(2,2)·T(−1,−1).

Q14 **Answer: 1.16**

Concept: Worst-case stack: the gap is largest when the housing is largest and every spacer is smallest.

G_{max} = 100.10 − 3 × 32.98 = 100.10 − 98.94

G_{max} = **1.16 mm** (and G_{min} = 99.90 − 99.06 = 0.84 mm)

Exam tip: The tolerance of the gap equals the sum of all the tolerances in the chain: ±(0.10 + 3 × 0.02) = ±0.16 mm about the nominal gap of 1.00 mm.

Q15 **Answer: (A) creates a new process plan from part features using decision logic, formulae and a manufacturing knowledge base, without starting from a stored plan**

Concept: CAPP comes in two types: variant (retrieval) and generative.

The variant approach classifies the part with a GT code, retrieves the standard plan of its family and modifies it. The generative approach synthesises the plan from scratch using decision tables, rules and process data: **(A)**.

Exam tip: Opitz and MICLASS are GT coding systems used to form part families for variant CAPP and cellular manufacturing.

What next?

Day 62 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 63/90: Engineering Mathematics, Probability & Statistics. Comment **GATE90** on our reel to get all links.
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